

AI-901 Cheat Sheet

Microsoft Azure AI Fundamentals · straight-to-the-point exam prep

Responsible AI · 25%

General AI use cases · 50%

Foundry / Azure OpenAI · 25%

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■ Straight to the point — the concepts, service picks and traps that decide your AI-901 pass. No filler.

Exam at a glance

700 / 1000

Passing score

40-60

Questions

~45-60 min

Duration

40-45% / 55-60%

Exam domains

Exam: AI-901 · Microsoft Azure AI Fundamentals — replaces AI-900 (retires 30 June 2026).

Domain 1 — Identify AI Concepts & Capabilities · 40-45%

Domain 2 — Implement AI Solutions by Using Microsoft Foundry · 55-60%

1 · Responsible AI — the six principles

Principle	What it means	Azure feature
Fairness	Equitable across groups	Fairness assessment / balanced data
Reliability & safety	Consistent & safe under stress	Content filters + evaluation
Privacy & security	Protects data end to end	Managed identities / private endpoints
Inclusiveness	Works for all abilities & languages	Accessibility & multilingual
Transparency	Decisions are understandable	Model cards
Accountability	Humans stay responsible	Audit logs + abuse monitoring

⚠ **Exam trap:** Fairness = the outcome is equitable; Transparency = the decision is explainable. A scenario names the principle — then often the Azure feature that upholds it.

2 · General AI use cases — pick the service ★ **Most tested**

Service	Use it for
Content Understanding (build an analyzer)	Free-form / multimodal extraction across docs, images, audio, video
Content Understanding - Read	OCR text from a page or image
Content Understanding - Layout	Text, tables, selection marks, structure
Document Intelligence prebuilt	Known forms: invoices, receipts, IDs, business cards
Azure Speech	Speech-to-text and text-to-speech
Azure Language	Sentiment, entities, PII, summarization
Azure OpenAI / Foundry models	Generate, reason, or understand an image

Analyzer: describe the fields you want in plain language; an LLM reasons over the content and returns structured JSON (prebuilt or custom).

⚠ **Exam trap:** Multimodal or free-form → Content Understanding. Known structured form → Document Intelligence prebuilt. No classic ML theory (supervised / unsupervised / regression / classification). Azure AI Vision – Image Analysis is retiring → use Content Understanding or a multimodal model.

3 · Foundry / Azure OpenAI

Concept	Key fact
Hub vs Project	Hub = governance / security / quota container. Project = where you build (deployments, agents, data, evals). One hub → many projects.
Portal map	Model catalog → Model card → Deployments → Playground → Agents → Evaluation
Foundry Tools	Speech, Language, Content Understanding & Document Intelligence exposed as tools inside Foundry

⚠ **Exam trap:** Deployment name (what your code calls) vs model name (e.g. gpt-4o) vs endpoint (resource URL + key) — don't mix them up.

Parameter	Effect	Exam cue
temperature	Low = deterministic / repeatable; high = creative	'consistent, repeatable output' → low temp
top_p	Nucleus sampling — tune one, not both	distractor: 'set both to 1 for accuracy'
max_tokens	Caps response length (prompt + completion)	'response got cut off' → raise it

Prompt role	Holds
system	Persona, tone, rules & guardrails for the whole conversation
user	The end-user's request this turn
assistant	Prior model replies — conversational memory

⚠ **Exam trap:** Behavioural rules ('be formal', 'stay on topic') belong in the SYSTEM prompt — not the user prompt, not deployment settings.

Concept	Key fact
Foundry SDK (read it)	azure-ai-projects + openai · auth with DefaultAzureCredential (no keys in code) · call the deployment name · read choices[0].message.content
Agent	model + instructions + tools + threads + runs. Tools: file search, code interpreter, function calling.
Agent vs chat call	Tools / state / multi-step → agent. One stateless answer → chat completion.

4 · Traps & X-vs-Y — memorize

X vs Y	What each side means
Hub vs Project	Hub = top-level governance, security & quota container. Project = where you build — deployments, agents, data, evaluations.
Deployment vs model vs endpoint	Deployment = the custom name your code calls. Model = the base model (e.g. gpt-4o). Endpoint = the resource URL + key.
Temperature vs top_p	temperature = randomness via sampling temperature. top_p = randomness via nucleus sampling. Tune one, not both.
System vs user prompt	system = persona, rules & guardrails for the whole chat. user = the end-user's request this turn.
Agent vs chat call	Agent = tools + state + multi-step actions. Chat completion = a single, stateless response.
Content Understanding vs Document Intelligence	Content Understanding = LLM analyzer for multimodal / free-form input. Document Intelligence = deterministic extraction from known forms.
Speech recognition vs synthesis	Recognition = speech-to-text (transcribe audio). Synthesis = text-to-speech (speak text).

✓ **Final strategy:** Bank Responsible AI + service picks (easy points). Spend real study time on Foundry — it's the biggest, hardest slice. Know the traps cold, and review Speech / Audio SDK. Skip classic ML theory.